$\Rightarrow\ (p-q)(p^2+q^2-pq)=(p-q)\{(p-q)^2-xpq\}$

$\{\because a^3-b^3=(a-b)(a^2-ab+b^2)\}$

by cancelling same terms of both sides

$\Rightarrow\ p^2+q^2-pq=p^2+q^2-2pq-xpq$

$\{(a-b^2)=a^2+b^2-2ab\}$

$\Rightarrow pq = -xpq$

$\Rightarrow x = -1$

**367.** 2361*48 will be divisible by 9 if the sum of the digits of the given number is divisible by 9.
2 + 3 + 6 + 1 + * + 4 + 8 i.e. (24 + *) is divisible by 9.
Clearly, * = 3 because 27 is divisible by 9.

**368.** Time taken by A to complete colouring a picture = $\frac{3}{4}$ hours
= 0.75 hours

Time taken by B to complete colouring a picture $\frac{7}{12}$ = hours
= 0.58 hours

Time taken by C to complete colouring a picture $\frac{5}{8}$ = hours
= 0.625 hours

Time taken by D to complete colouring a picture $\frac{6}{7}$ = hours = 0.85 hours

Hence, B took least time in colouring the picture.

**369.** Let number be $a$
According question, $\frac{4}{5}$ of $a$ – 45% of $a$ = 56

$\frac{4}{5}a-\frac{45}{100}\times a=56$

$\frac{35a}{100}=56\Rightarrow a=\frac{100\times 56}{35}$

$\Rightarrow a = 160$

= 65% of $a$ = $\frac{65}{100}\times 160=104$

**370.** Given $x+y:y+z:z+x=6:7:8$

$\frac{x+y}{6}=\frac{y+z}{7}=\frac{z+x}{8}=a$

(let)

$\Rightarrow x + y = 6a$ .... (i)

$y + z = 7a$ …(ii)

$z + x = 8a$ …(iii)

On adding all three equations

$x+y+y+z+z+x=6a+7a+8a$

$\Rightarrow\ 2(x+y+z)=21a$

$\Rightarrow 2\times 14=21a\ [\because x+y+z=14]$

$\therefore\ a=\frac{2\times 14}{21}=\frac{4}{3}$

$\therefore\ x+y=6a=6\times\frac{4}{3}=8$

$\therefore\ z=(x+y+z)-(x+y)$

= 14 – 8 = 6

**371.** A megabyte is 1.048, 576 bytes or 1,024 kilobytes. It conveniently expression the binary multiples inherent in digital computer memory architectures. However, megabyte architectures. However, megabyte is also taken to mean 1000 × 1024 (1024000) bytes.

**372.** The first 3-digit number which is divisible by 9 is 108 and last three digit number which is divisible by 9 is 999.
So, we have an AP with $a$ = 108, $d$ = 9 and $a_n$ = 999

$\therefore a_n = a + (n - 1)d$

$\Rightarrow 999 = 108 + (n - 1)\ 9$

$\Rightarrow 999 - 108 = (n - 1)\ 9$

$\Rightarrow 891 = (n - 1)\ 9$

$\Rightarrow (n - 1) = \frac{891}{9} = 99$

$\Rightarrow n + 99 + 1 = 100$

**373.** Let, the two consecutive even numbers are $a$ and $(a + 2)$, respectively
According to question,

$a^2+(a+2)^2=1060\ \{\because (a+b)^2=a^2+b^2+2ab\}$

$\Rightarrow a^2 + a^2 + 4 + 4a = 1060$

$\Rightarrow 2a^2+4a-1056=0$

$\Rightarrow a^2+2a-528=0$

$\Rightarrow a^2+24a-22a-528=0$

$\Rightarrow a(a+24)-22(a+24)=0$

$\Rightarrow (a-22)(a+24)=0$

$\Rightarrow a = -24, 22$

**374.** The data in both the statements I and II are not sufficient to answer the question.

**375.** Given $n = p_1^{x_1}\, p_2^{x_2}\, p_3^{x_3}$ where $p_1, p_2, p_3$ are distinct prime factors
Number of prime factors form = $(x_1 \times x_2 \times x_3) = x_1\, x_2\, x_3$
Hence, option (b) is correct

**376.** 11, 111, 1111, 11111, …..
Let $m$ = 2 ⇒ 4 × 2 + 3 = 11
$m$ = 27 ⇒ 4 × 27 + 3 = 111
Each number can be expressed in the form (4$m$ + 3) where $m$ is a natural number
Hence, statement 1 is only correct.

**377.** Maximum daily wages of an officers
= H.C.F. of ₹ 4,956; and ₹ 3,894
= ₹ 354
Illustration:

$$\begin{array}{r} 3894\overline{)4956}\,(1 \\ 3894 \\ \hline 1062\overline{)3894}\,(3 \\ 3186 \\ \hline 708\overline{)1062}\,(1 \\ 708 \\ \hline 354\overline{)708}\,(2 \\ 708 \\ \hline \times \end{array}$$